ROYAL FITNESS CLUB

Body Transformation Blueprint

MODULE 3

Muscle Building for Beginners

Duration	55 minutes
Course Code	CS_B3
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: The Foundations of Hypertrophy

What Makes a Muscle Actually Grow?

Skeletal muscle hypertrophy (growth in size) occurs when muscle protein synthesis (MPS) consistently exceeds muscle protein breakdown (MPB) over days, weeks, and months. This net positive protein balance results in larger muscle fibres with more contractile proteins (actin and myosin) and, ultimately, a larger muscle belly.

Three primary mechanical and metabolic stimuli drive hypertrophy, and understanding their relative contribution guides optimal programme design:

Mechanism	Contribution	Training Application	Evidence Strength
Mechanical tension (primary driver)	~60–70% of hypertrophic signal	Heavy compound lifts (>60% 1RM), full ROM, controlled eccentrics, stretched position loading	Very strong — Schoenfeld et al., meta-analyses
Metabolic stress (pump)	~20–25%	High-rep sets (15–30 reps), short rest (<90s), blood flow restriction, supersets, drop sets	Moderate — mechanism debated but effect real
Muscle damage	~10–15%	Novel exercises, slow eccentrics (4–5s), stretched position under load (RDL, deep squat, flyes)	Moderate — may indicate satellite cell activation

Myofibrillar vs Sarcoplasmic Hypertrophy

Type	What Grows	Training Stimulus	Outcome	Look
Myofibrillar	Actin and myosin protein content — the contractile machinery	Heavy loads (>75% 1RM), lower reps (3–8)	Denser, stronger muscle — 'hard' look	Strength athlete physique
Sarcoplasmic	Glycogen, water, organelles in the muscle cell	Moderate loads, higher reps (8–20), pump work	Larger muscle — greater size without proportional strength gain	Bodybuilder 'full' look

Both types of hypertrophy occur with resistance training. The relative proportion depends on rep range and loading scheme. Most evidence suggests a spectrum of 6–20 reps produces optimal total hypertrophy when combined with adequate volume and progressive overload.

SECTION 2: Volume — The Most Important Training Variable for Hypertrophy

How Many Sets Do You Need?

Training volume — typically expressed as weekly sets per muscle group — is the strongest predictor of hypertrophy in research and practice. More volume produces more hypertrophy up to a recovery-limited ceiling ("maximum adaptive volume"), beyond which additional sets become junk volume.

Volume Category	Sets/Muscle/Week	Who It's For	Note
Minimum Effective Volume (MEV)	10–12 sets	Beginners; maintenance	Sufficient for near-maximal beginner response
Moderate Volume	12–18 sets	Intermediate; most recreational training	Good balance of stimulus and recovery

High Volume	18–25 sets	Advanced; well-recovered; periodised	Requires programmed deloads; high-protein diet
Maximum Adaptive Volume (MAV)	25–35 sets	Specialisation phases only; very advanced	Beyond this = recovery deficit

Progressive volume: Start at MEV (10–12 sets/week/muscle) and add 2 sets per muscle per week across a 4–6 week mesocycle, then deload and restart slightly above previous MEV. This is "volume progression" — a key element of long-term hypertrophy planning.

Hard sets only: Volume only counts as "hard" sets — taken to within 3 reps of failure (RPE 7+). Warm-up sets, feeder sets at RPE 5 do not contribute meaningfully to the hypertrophic stimulus. Quality over quantity.

SECTION 3: Exercise Selection for Maximum Hypertrophy

The Best Exercises for Each Muscle Group

Muscle Group	Primary Exercise	Secondary	Isolation	Notes
Chest	Barbell bench press, incline DB press	Dip, cable crossover	Pec deck, cable fly	Incline hits clavicular head better; stretch-focused at b
Back (width)	Pull-up, lat pulldown	Cable pullover	Straight-arm pulldown	Full shoulder extension → full lat activation
Back (thickness)	Barbell row, cable row	DB row, machine row	Face pull (rear delt/mid trap)	Elbow close to body = lat; elbow out = rhomboid/tr
Shoulders	Barbell OHP, DB press	Arnold press	Lateral raise (15–30 reps)	Lateral raise best at 30° scaption; avoid shruggin
Biceps	Incline DB curl (stretch)	Barbell curl, preacher curl	Concentration curl, cable curl	Stretch position (incline, cable from high) superior for hyp
Triceps	Close-grip bench, dip	Skull crusher	Pushdown (pronated), overhead cable ext.	Long head (back of tricep) best hit with arm overhe
Quadriceps	Back squat, leg press	Front squat, hack squat	Leg extension (for terminal knee ext.)	Knee over toe = more quad; hip hinge = more glu
Hamstrings	Romanian deadlift, Nordic curl	Leg curl, good morning	Seated leg curl	Stretched position (at hip AND knee) = most damage +
Glutes	Hip thrust, Bulgarian split squat	Squat, deadlift	Kickback, abduction	Hip thrust activates glute max 30%+ more than squ
Calves	Standing calf raise	Donkey calf raise	Seated calf raise (soleus)	Full ROM: heel below level of step; hold stretch at bo

SECTION 4: Anabolic Hormones and Muscle Growth

The Hormonal Environment for Muscle Building

Hormone	Role in Hypertrophy	How Training Affects It	Optimise By
Testosterone	Androgen receptor binding → increased gene transcription [illegible]	[illegible] compound lifts (>75% 1RM) [illegible] MPS, IGF-1 [illegible]	[illegible] dietary fat [illegible] (>15%) [illegible] sleep 8h [illegible]
Growth Hormone (GH)	IGF-1 production (liver); fat mobilisation; collagen synthesis	Exercise (especially heavy + metabolic) → GH pulse	Deep sleep (largest pulse during N3 sleep); avoid alcohol pre-sleep
IGF-1 (Insulin-like Growth Factor 1)	Satellite cell activation; protein synthesis via PI3K/Akt/mTOR pathway	Exercise and high-protein diet increase liver IGF-1 production	Resistance training; adequate calorie

Insulin	GLUT4 → amino acid uptake, synergistic with mTOR for MPS	Post-workout carbohydrate + protein	Elevation during muscle...

The "hormone spike" myth: Acute post-exercise testosterone and GH spikes from training are real but do NOT directly cause hypertrophy. The local mechanical stimulus (mTORC1 in the trained muscle) is the primary driver. Hormone spikes are correlated with, not causal of, hypertrophy — explaining why upper body exercises cause systemic testosterone spikes but don't produce upper body growth proportional to the spike.

Module 3 (CS_B3) Key Takeaways
• Hypertrophy requires MPS > MPB consistently; mechanical tension is the primary stimulus (~60–70%)
• Volume (sets/week/muscle) is the most important training variable for hypertrophy; start at 10–12 hard sets
• Progress volume over each mesocycle (+2 sets/week), then deload; begin next cycle above previous MEV
• Stretch-position loading (incline curls, RDL, full ROM squats) produces greatest muscle damage and growth signal
• Hip thrust produces 30%+ greater glute max activation than squat — include both for complete development
• Anabolic hormones (testosterone, GH, IGF-1) create a permissive environment but local mTORC1 is the primary driver

1. Explain the three mechanisms of hypertrophy and their relative contribution.

2. What is the difference between MEV (minimum effective volume) and MAV (maximum adaptive volume)?

3. Why does the incline dumbbell curl produce more hypertrophy than a standard standing barbell curl?

4. Describe volume progression across a 6-week mesocycle starting at 12 sets for biceps.

5. What is the "hormone spike" myth and why doesn't the acute testosterone response from squats build upper body muscle?